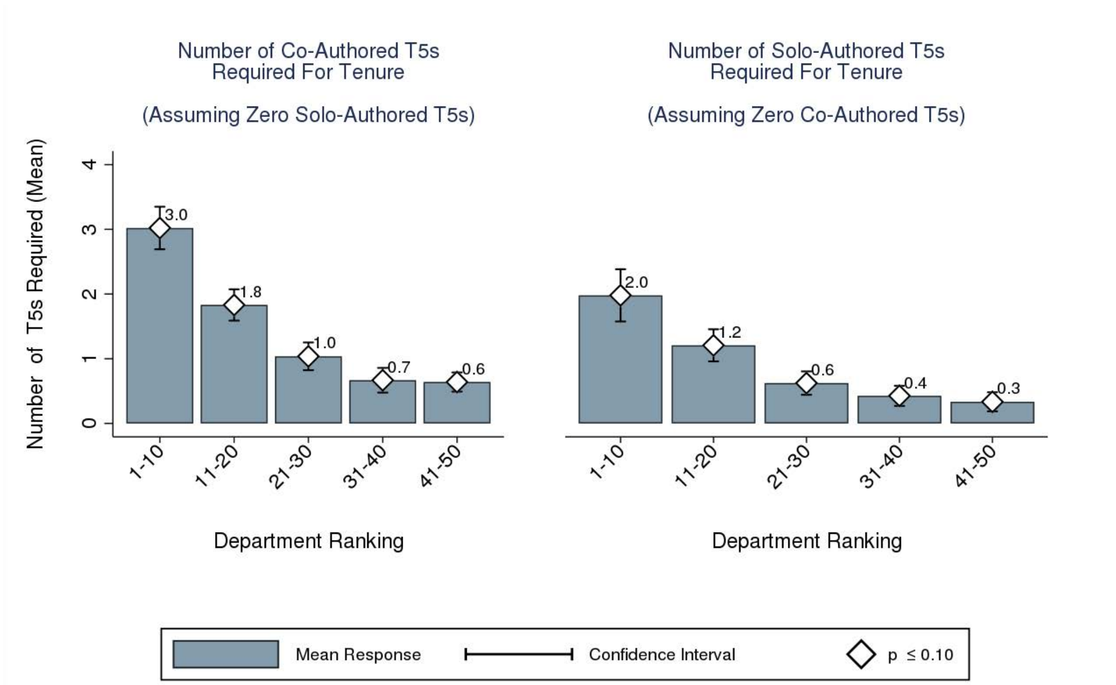
on judging candidates' research output, and given that external reviewers likely work in departments that are in the same class as the candidate's department (with similar levels of T5 emphasis in research evaluation), it is possible that external reviewers put as large an emphasis on a candidate's quantity of T5 publications as reviewers who are internal to the candidate's department. Indeed, it is not unusual for letter writers to focus on the number of T5 articles published or in the pipeline for a prospective candidate. Such dependence of external letters on the quantity of T5 publications would compound the pressure faced by junior faculty to publish in the T5.

Figure 12: Minimum Number of T5 Publications Required for Tenure



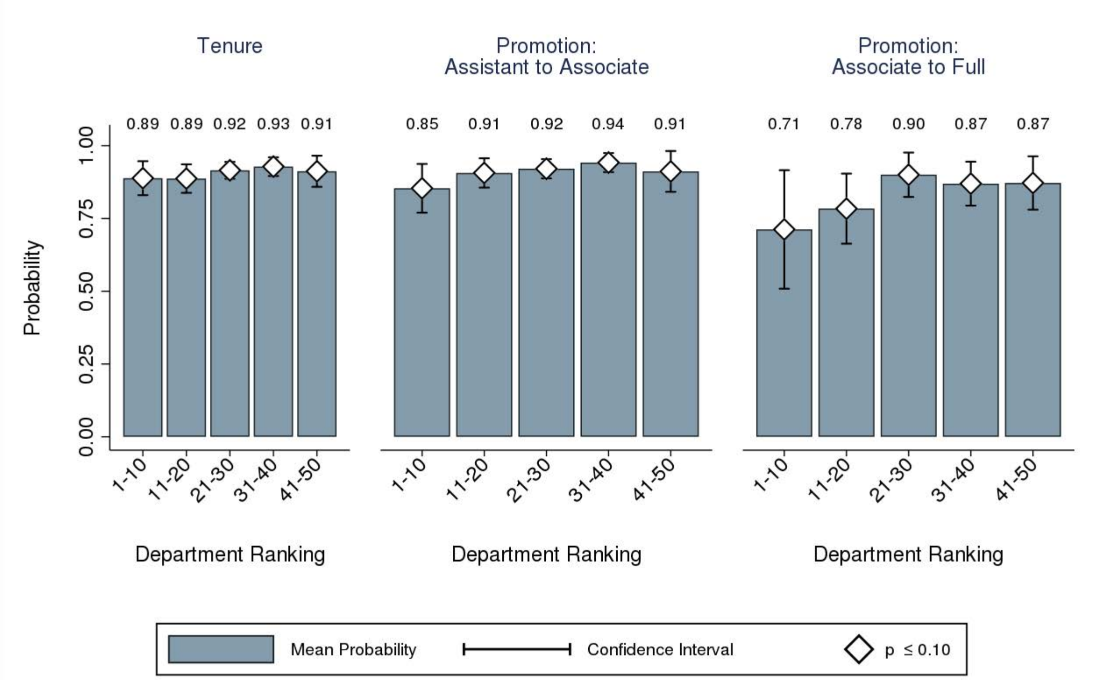
Note: This figure summarizes respondents' perceptions about the number of T5 publications that are required to obtain tenure in their department. The bars present mean responses for each performance area. White diamonds indicate that the responses were significantly different than zero at the 10% level.

Non-T5 publications receive the third-highest mean rank across all levels of career advancement. However, the rankings for both external letters and non-T5 publications are only significantly different than the rankings for citations when we consider tenure and promotion to associate professor. The Wilcoxon tests presented in Online Appendix Table O-A58 fail to reject the null that the ranking distributions for external letters and non-T5 publications are equal to the distribution for citations for promotions to full professor. The remaining performance areas receive the four lowest mean ranks across all career advancement types. Teaching performance and success in securing grants receive rankings that are not significantly different from each other for any type of career advancement. Books and chapters in books are ranked last for all levels of career advancement. Long-term integrated bodies of research are valued much less than focused T5 articles.

These survey results offer important evidence on the large influence of T5 publications on tenure and promotion practices. However, they do not shed light on whether the difference in influence between T5 and non-T5 publications is merely a reflection of differences in article impact and quality between these outlets, or whether the T5's influence also operates through channels that are independent of article impact and quality. Figure 13 presents some evidence that answers this question. The figure summarizes responses to a survey question that asks respondents to compare the probabilities of receiving tenure and promotion associated with publishing in T5 and non-T5 journals, fixing the quality of the publications in question to be equal. Specifically, the question presents respondents with a thought experiment wherein respondents are asked to imagine a scenario where their departments must decide to tenure and/or promote one out of two candidates. The respondents are asked to assume that the two candidates are identical in every respect, with the exception that one candidate has published all of their articles in T5 journals whereas the other candidate has published the same number of articles of equal quality in non-T5 journals. The respondents are then asked to report the probability that the candidate with T5 publications receives tenure and/or promotion instead of the candidate with non-T5 publications. In a scenario where the T5 influence operates solely through differences in article impact and quality, both the T5 and non-T5 candidate would be expected to receive tenure and/or promotion with a probability of 0.5. Any deviation from 0.5 in favor of the T5 candidate indicates that the T5 influences tenure and/or promotion decisions through channels that are independent of article quality.

The results plotted in Figure 13 reveal large and statistically significant deviations

Figure 13: Probability That Candidate with T5 Publications Receives Tenure or Promotion Instead of Candidate with non-T5 Publications, *ceteris paribus*



Note: This figure summarizes respondents' perceptions about the probability that a candidate with T5s is granted tenure or promotion by the respondent's department instead of a candidate with non-T5s, *ceteris paribus*. Responses are summarized by type of career advancement: tenure receipt, promotion to assistant professor, and promotion to associate professor. The bars present mean responses for each performance area. White diamonds indicate that the mean response is significantly different than 50% at the 10% level.

from 0.5 in favor of the candidate with more T5s. The deviations exist across department rank groupings, and for all three levels of career advancement: tenure receipt, promotion to assistant professor, and promotion to associate professor. The figure plots the mean response by department rank group and level of career advancement. For tenure decisions, the mean response is 0.89 or higher across all department rank groups. Thus, on average, junior faculty at the top 50 departments believe that their department would award tenure to the T5 candidate instead of the non-T5 candidate at least 89 times out of 100. The mean reported probability rises as one considers lower ranked departments, with its value peaking at 0.93 for departments ranked 31–40. The reported probabilities are similarly high for promotions to associate professor. Mean reported probabilities are lower for promotions to full professor, and exhibit higher variation. However, the means continue to remain significantly different than 0.5 at the 10% level.

These results reveal that there exists a widely-held belief among junior faculty at the Top 50 departments that the same quantity and quality of articles will yield rewards at vastly different rates based on whether their articles are published in T5 or non-T5 journals. Junior faculty form expectations based on past decisions, and past decisions are clearly biased in favor of T5 publishing (see Figure 5). For rational career-oriented economists who prioritize tenure and career advancement, given the current incentives academic careers should be little more than quests for publication in the T5.

4 The T5 as a Filter of Quality

The analysis of Section 2 establishes the strong relationship between tenure decisions in the top 35 departments and T5 publications. The analysis of Section 3 shows that junior faculty are acutely aware of the power of the T5. The analysis in this section evaluates quality of the T5 as a filter of research influence and quality. Using citations as a proxy for influence, Section 4.1 compares citation distributions of individual journals against the citation distribution of T5 journals as a group. Section 4.2 compares journals with respect to the share of the most influential papers that have been published by T5 and non-T5 journals. Section 4.3 compares T5 and non-T5 journals based on Impact Factors. Section 4.4 examines the publishing behavior of influential economists from 14 major fields of economics.

4.1 Comparison of Citations Between T5 and Non-T5 Journals

This section compares cumulative citation counts (measured as of 2018) of articles published in the T5 and those published in 25 non-T5 journals over the ten year period 2000–2010. We reluctantly follow the literature in using citation counts as a valid measure of productivity. However, it is obvious that citation counts are flawed measures of productivity. It is very likely that, following convention, authors are more likely to cite T5 papers even when comparable or superior non-T5 papers are available. Given current practices, the appeal to a T5-certified paper strengthens a reference in the eyes of many readers. Academic productivity is an elusive concept. In the absence of a better measure, we reply on a flawed measurer. Our analysis shows large intra-journal heterogeneity and inter-journal overlap in the quality of published articles across T5 and non-T5 journals. Combined with our analysis in Section 4.2 which identifies non-T5 journals that produce as many, if not more, influential articles than the T5, the findings of this section show that whether an article is published in the T5 is a poor predictor of the article's actual quality.

The comparisons in this section build on the analysis of Hamermesh (2018), who compares citations in the T5 journals, with citations in the *Review of Economics and Statistics* and the *Economic Journal*. We extend his analysis by expanding the set of non-T5 journals considered to 25, and by analyzing articles published in a wider and more recent time frame (2000–2010 in our analysis vs. 1974-75 and 2007-08 in Hamermesh (2018)).[54] Our results confirm his findings. There are large intra-T5 variation in citations and large overlaps in citations between papers published in the T5, and those published in *ReStat* and *EJ*. Our use of the expanded journal comparison set helps identify six additional non-T5 economics journals that share at least as large a citation overlap with the T5 as *EJ*. We conclude the analysis by comparing the overlap between non-T5 journals and different subsets of T5 journals. We find that the comparability between T5 and non-T5 publications greatly increases when one focuses on the lesser-cited T5 journals. As a case in point, the median-cited *ReStat* article ranks in the 38^{th} percentile of year-adjusted citations among *all* T5 publications, but attains a rank of the 58^{th} percentile when compared to *ReStud* alone. These comparisons illustrate the large heterogeneity in influence among the journals that comprise the T5.

We note at the outset that for want of a better measure, our comparisons of journal and article quality rely on citations. One concern with using citations is that it could

[54]Our chosen time frame necessarily excludes any analysis of the impact of the new *AEA* applied journals, which started publication in 2009.